NEET Comprehensive Revision Guide

High-Yield Concepts, Core Formulas & Essential NCERT Highlights

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Strict NCERT & NMC Syllabus-Aligned Comprehensive Master Notes

Curriculum & Topic Coverage Overview

This revision handbook serves as an exhaustive, fast-track reference guide covering 10 high-yield chapters across all four core NEET subjects. Every module synthesizes standard NCERT theory, foundational mechanics, key derivations, reaction pathways, and high-probability past-year question patterns.

- **Physics**
  - **Laws of Motion** — Newton's Laws, Momentum, Impulse, Static & Kinetic Friction, Circular Motion & Banking Dynamics.
  - **Work, Energy and Power** — Work-Energy Theorem, Conservative Forces, Spring & Gravitational Potential Energy, Vertical Circular Motion, Collisions.
  - **Centre of Mass and System of Particles** — Centre of Mass of Standard Rigid Bodies, Conservation of Linear Momentum, Rotational Dynamics, Moment of Inertia, Rolling Motion.
- **Chemistry**
  - **Electrochemistry** — Galvanic Cells, Nernst Equation, Gibbs Free Energy, Electrolytic Conductivity, Kohlrausch's Law, Faraday's Laws, Batteries & Fuel Cells.
  - **Chemical Kinetics** — Rate Law, Integrated Rate Equations (Zero & First Order), Half-Life Relations, Arrhenius Equation, Activation Energy, Collision Theory.
  - **Structure of Atom** — Bohr's Atomic Model, Rydberg Formula, de Broglie Wavelength, Heisenberg Uncertainty Principle, Quantum Numbers, Electronic Configuration Principles.
- **Biology — Botany**
  - **Plant Kingdom** — Algae (Chlorophyceae, Phaeophyceae, Rhodophyceae), Bryophytes, Pteridophytes (Heterospory & Seed Habit Precursor), Gymnosperms, Angiosperms, Alternation of Generations.
  - **Morphology of Flowering Plants** — Modifications of Roots, Stems, and Leaves; Inflorescence; Floral Architecture; Aestivation; Placentation; Key Families (Fabaceae, Solanaceae, Liliaceae).
- **Biology — Zoology**
  - **Excretory Products & Their Elimination** — Ammonotelism, Ureotelism, Uricotelism; Nephron Anatomy; Ultrafiltration; Counter-Current Mechanism; Hormonal Regulation (JGA/RAAS, ADH, ANF); Renal Disorders.
  - **Locomotion & Movement** — Muscle Ultrastructure (Actin, Myosin, Sarcomere); Sliding Filament Mechanism; Red vs White Fibres; Human Skeleton (206 Bones); Joints; Musculoskeletal Disorders.

1. Physics: Core Principles, Mechanics & Rotational Dynamics

1.1 1. Laws of Motion

1.1.1 Newton's Laws & Momentum Conservation

- **First Law (Law of Inertia):** An object remains at rest or in uniform motion along a straight line unless acted upon by an external net force. Inertia depends directly on inertial mass.
- **Second Law:** The time rate of change of linear momentum is directly proportional to the applied force:
$$\vec{F}_{\text{net}} = \frac{d\vec{p}}{dt} = \frac{d(m\vec{v})}{dt} = m\vec{a} \quad \text{(for constant mass } m\text{)}$$
- **Third Law:** To every action, there is always an equal and opposite reaction ($\vec{F}_{AB} = -\vec{F}_{BA}$). Action and reaction act simultaneously on **two different bodies**, never cancelling each other out.
- **Impulse ($\vec{I}$):** Product of average force and duration of action, equal to the net change in momentum:
$$\vec{I} = \int_{t_1}^{t_2} \vec{F}\, dt = \vec{F}_{\text{avg}}\Delta t = \Delta\vec{p} = m(\vec{v} - \vec{u})$$
Graphical representation: The area bounded under the Force–Time (F–t) curve represents the total impulse $\vec{I}$.
- **Conservation of Linear Momentum:** In an isolated system where external force is zero ($\vec{F}_{\text{ext}} = 0$):
$$\sum \vec{p}_{\text{initial}} = \sum \vec{p}_{\text{final}} \implies m_1\vec{u}_1 + m_2\vec{u}_2 = m_1\vec{v}_1 + m_2\vec{v}_2$$
Recoil Velocity of Gun: $v_{\text{gun}} = -\frac{m_{\text{bullet}}}{M_{\text{gun}}} v_{\text{bullet}}$.

1.1.2 Friction Dynamics

Friction is an electromagnetic contact force opposing relative impending or actual motion between contacting surfaces.

- **Static Friction (f_s):** Self-adjusting force ($0 \le f_s \le f_{s,\max}$).
$$f_{s,\max} = \mu_s N$$
where μ_s is the coefficient of static friction, and N is the normal contact force.
- **Kinetic Friction (f_k):** Nearly independent of relative sliding velocity:
$$f_k = \mu_k N \quad (\mu_k < \mu_s)$$
- **Angle of Friction (λ):** Angle between total contact force $\vec{R} = \vec{N} + \vec{f}_s$ and the normal $\vec{N}$:
$$\tan\lambda = \frac{f_{s,\max}}{N} = \mu_s \implies \lambda = \tan^{-1}(\mu_s)$$
- **Angle of Repose (θ):** Minimum incline angle at which a body on an inclined plane begins to slide down:
$$mg\sin\theta = \mu_s mg\cos\theta \implies \tan\theta = \mu_s \implies \theta = \lambda$$

Connected Systems & Elevator Dynamics

1. Apparent Weight in an Elevator of mass m:

- Upward acceleration $+a$: $N = m(g + a)$ (Heavier sensation)
- Downward acceleration $-a$: $N = m(g - a)$ (Lighter sensation)
- Free fall ($a = g$): $N = m(g - g) = 0$ (Weightlessness)

2. Atwood Machine (Two masses $m_2 > m_1$ over a frictionless light pulley):

$$a = \frac{m_2 - m_1}{m_1 + m_2} g, \qquad T = \frac{2m_1 m_2}{m_1 + m_2} g$$

1.1.3 Circular Motion Dynamics & Banking of Curves

When a body negotiates a curved path of radius r at speed v, centripetal acceleration $a_c = \frac{v^2}{r}$ is required.

- **Level Unbanked Road:** Centripetal force supplied entirely by static friction:

$$\frac{mv^2}{r} \leq \mu_s N = \mu_s mg \implies v_{\max} = \sqrt{\mu_s g r}$$

- **Banked Road without Friction (Optimum / Design Speed v_0):**

$$N\sin\theta = \frac{mv_0^2}{r}, \quad N\cos\theta = mg \implies \tan\theta = \frac{v_0^2}{rg} \implies v_0 = \sqrt{rg\tan\theta}$$

- **Banked Road with Friction (μ_s):**

$$v_{\max} = \sqrt{rg\left(\frac{\mu_s + \tan\theta}{1 - \mu_s\tan\theta}\right)}, \qquad v_{\min} = \sqrt{rg\left(\frac{\tan\theta - \mu_s}{1 + \mu_s\tan\theta}\right)}$$

- **Bending of a Cyclist:** $\tan\theta = \frac{v^2}{rg}$ (measured from vertical).

1.2 2. Work, Energy and Power

1.2.1 Work Done & Work-Energy Theorem

- **Constant Force:** $W = \vec{F}\cdot\vec{s} = Fs\cos\theta$.
- **Variable Force in 1D:** $W = \int_{x_i}^{x_f} F(x)\,dx$ (Area under the F–x curve).
- **Work-Energy Theorem:** The total work done by all forces (conservative, non-conservative, and pseudo) equals the net change in kinetic energy:

$$W_{\text{net}} = W_{\text{conservative}} + W_{\text{non-conservative}} + W_{\text{pseudo}} = \Delta K = K_f - K_i = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$

1.2.2 Conservative Forces & Potential Energy (U)

Potential energy is defined strictly for conservative forces where work is path-independent:

$$F = -\frac{dU}{dx} \quad \left(\text{in 3D: } \vec{F} = -\nabla U = -\left[\frac{\partial U}{\partial x}\hat{i} + \frac{\partial U}{\partial y}\hat{j} + \frac{\partial U}{\partial z}\hat{k}\right]\right)$$

- **Gravitational PE:** $U(h) = mgh$ (for $h \ll R_{\text{Earth}}$).
- **Spring PE:** Restoring force $F = -kx \implies U = \frac{1}{2}kx^2$.
- **Equilibrium States:**
  - Stable Equilibrium: $\frac{dU}{dx} = 0$, $\frac{d^2U}{dx^2} > 0$ (Potential energy is minimum).
  - Unstable Equilibrium: $\frac{dU}{dx} = 0$, $\frac{d^2U}{dx^2} < 0$ (Potential energy is maximum).
  - Neutral Equilibrium: $\frac{dU}{dx} = 0$, $\frac{d^2U}{dx^2} = 0$ (U remains constant).

Vertical Circular Motion (String of Length R)

For a particle of mass m tied to a massless string of length R:

- Critical speed at bottom point (L): $v_L \geq \sqrt{5gR}$ (Minimum for complete loop).
- Critical speed at highest point (H): $v_H \geq \sqrt{gR}$ (String just stays taut, $T_H = 0$).
- Speed at horizontal point (M): $v_M \geq \sqrt{3gR}$.
- Tension difference between bottom and top: $T_L - T_H = 6mg$ (Constant for any looping speed).

1.2.3 Collisions in One Dimension

- **Coefficient of Restitution (e):**

$$e = \frac{\text{Velocity of Separation}}{\text{Velocity of Approach}} = \frac{v_2 - v_1}{u_1 - u_2}$$

  - Perfectly Elastic Collision: $e = 1$ (Momentum & Kinetic Energy conserved).
  - Inelastic Collision: $0 < e < 1$ (Momentum conserved, Kinetic Energy lost).
  - Perfectly Inelastic Collision: $e = 0$ (Bodies stick together, max KE loss).

- **Elastic Head-on Collision Velocities ($e = 1$):**

$$v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) u_1 + \left(\frac{2m_2}{m_1 + m_2}\right) u_2, \qquad v_2 = \left(\frac{2m_1}{m_1 + m_2}\right) u_1 + \left(\frac{m_2 - m_1}{m_1 + m_2}\right) u_2$$

  Special Case ($m_1 = m_2$): Velocities are completely exchanged ($v_1 = u_2, v_2 = u_1$).

- **Loss of Kinetic Energy in Inelastic Collision ($0 \leq e \leq 1$):**

$$\Delta K = \frac{1}{2}\left(\frac{m_1 m_2}{m_1 + m_2}\right)(u_1 - u_2)^2(1 - e^2)$$

1.3 3. Centre of Mass and System of Particles

1.3.1 Centre of Mass (CM) Coordinates

- **Discrete System:** $\vec{r}_{\text{cm}} = \frac{\sum m_i \vec{r}_i}{\sum m_i} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2 + \ldots}{M}$.
- **Two-Particle System:** Distance of CM from m_1 on separation line r: $r_1 = \left(\frac{m_2}{m_1 + m_2}\right) r$.
- **Continuous Mass Distribution:** $\vec{r}_{\text{cm}} = \frac{1}{M}\int \vec{r}\, dm$.

Rigid Body Geometry	Symmetry Axis / Reference	CM Location Coordinate (y_{cm})
Uniform Semicircular Ring	Along symmetry axis from centre	$y_{\text{cm}} = \frac{2R}{\pi} \approx 0.637R$
Uniform Semicircular Disc	Along symmetry axis from centre	$y_{\text{cm}} = \frac{4R}{3\pi} \approx 0.424R$
Uniform Hemispherical Shell	Along central radius from base	$y_{\text{cm}} = \frac{R}{2} = 0.5R$
Uniform Solid Hemisphere	Along central radius from base	$y_{\text{cm}} = \frac{3R}{8} = 0.375R$
Uniform Hollow Right Cone	Along altitude from base	$y_{\text{cm}} = \frac{h}{3}$
Uniform Solid Right Cone	Along altitude from base	$y_{\text{cm}} = \frac{h}{4}$

Table 1: Coordinates of Centre of Mass for Standard Symmetrical Bodies

1.3.2 Rotational Kinematics & Dynamics

- **Torque ($\vec{\tau}$):** $\vec{\tau} = \vec{r} \times \vec{F} = I\vec{\alpha} = \frac{d\vec{L}}{dt}$.
- **Angular Momentum ($\vec{L}$):** $\vec{L} = \vec{r} \times \vec{p} = I\vec{\omega}$.
- **Law of Conservation of Angular Momentum:** If $\vec{\tau}_{\text{ext}} = 0 \implies \vec{L} = \text{constant} \implies I_1\omega_1 = I_2\omega_2$.
- **Moment of Inertia Theorems:**
  - Parallel Axis Theorem: $I = I_{\text{cm}} + Md^2$ (Applicable to any 3D rigid body).
  - Perpendicular Axis Theorem: $I_z = I_x + I_y$ (**Only** valid for 2D laminar planar bodies).

Moments of Inertia (I) of Standard Bodies

- Thin Circular Ring (axis perpendicular through centre): $I = MR^2$; along diameter: $I = \frac{1}{2}MR^2$.
- Uniform Circular Disc (perpendicular through centre): $I = \frac{1}{2}MR^2$; along diameter: $I = \frac{1}{4}MR^2$.
- Solid Sphere (about any diameter): $I = \frac{2}{5}MR^2$; tangent: $I = \frac{7}{5}MR^2$.
- Hollow Sphere / Thin Spherical Shell: $I = \frac{2}{3}MR^2$; tangent: $I = \frac{5}{3}MR^2$.
- Thin Uniform Rod of length L (perpendicular through centre): $I = \frac{1}{12}ML^2$; through end: $I = \frac{1}{3}ML^2$.

1.3.3 Pure Rolling Motion on an Incline

For a body of mass M, radius R, and radius of gyration k rolling without slipping ($v_{\text{cm}} = R\omega$):

- **Total Kinetic Energy:**

$$K_{\text{total}} = K_{\text{trans}} + K_{\text{rot}} = \frac{1}{2}Mv_{\text{cm}}^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}Mv_{\text{cm}}^2\left(1 + \frac{k^2}{R^2}\right)$$

- **Linear Acceleration Down an Incline (θ):**

$$a = \frac{g\sin\theta}{1 + \frac{k^2}{R^2}}, \qquad v_{\text{bottom}} = \sqrt{\frac{2gh}{1 + \frac{k^2}{R^2}}}$$

- **Order of Acceleration / Speed ($a \propto \frac{1}{1+k^2/R^2}$):** Solid Sphere ($\frac{k^2}{R^2} = 0.4$) > Disc ($\frac{k^2}{R^2} = 0.5$) > Hollow Sphere ($\frac{k^2}{R^2} = 0.67$) > Ring ($\frac{k^2}{R^2} = 1.0$).

2. Chemistry: Physical Principles, Kinetics & Atomic Structure

2.1 4. Electrochemistry

2.1.1 Galvanic Cells & Nernst Equation

- **Standard EMF of Cell:** $E^\circ_{\text{cell}} = E^\circ_{\text{cathode (reduction)}} - E^\circ_{\text{anode (reduction)}}$.
- **Standard Hydrogen Electrode (SHE):** $\text{Pt}(s) \mid \text{H}_2(g, 1\text{ bar}) \mid \text{H}^+(aq, 1\text{ M})$, by convention assigned $E^\circ = 0.00$ V.
- **General Nernst Equation (at 298 K / 25°C):**

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{2.303RT}{nF}\log_{10} Q = E^\circ_{\text{cell}} - \frac{0.0591}{n}\log_{10} Q$$

  where n is moles of electrons transferred, and Q is the reaction quotient.
- **Equilibrium State ($E_{\text{cell}} = 0$, $Q = K_c$):**

$$E^\circ_{\text{cell}} = \frac{0.0591}{n}\log_{10} K_c \implies \log_{10} K_c = \frac{nE^\circ_{\text{cell}}}{0.0591}$$

- **Gibbs Free Energy & Work:**

$$\Delta G = -nFE_{\text{cell}}, \qquad \Delta G^\circ = -nFE^\circ_{\text{cell}} = -2.303RT\log_{10} K_c$$

  Criterion for spontaneity: $E_{\text{cell}} > 0 \iff \Delta G < 0$.

2.1.2 Conductance & Electrolytic Conduction

- **Conductivity (Specific Conductance, κ):** Inverse of resistivity (ρ):

$$\kappa = \frac{1}{\rho} = \frac{1}{R}\left(\frac{l}{A}\right) = G \cdot G^*$$

  where $G = \frac{1}{R}$ is conductance (Siemens, S), and $G^* = \frac{l}{A}$ is the **Cell Constant** (cm^{-1} or m^{-1}).
- **Molar Conductivity (Λ_m):**

$$\Lambda_m = \frac{\kappa \times 1000}{M} \quad \text{(Units: S cm}^2\text{ mol}^{-1}\text{, with } \kappa \text{ in S cm}^{-1}, M \text{ in molarity)}$$

- **Dilution Behavior:**
  - Conductivity (κ) **decreases** upon dilution because the number of conducting ions per unit volume decreases.
  - Molar conductivity (Λ_m) **increases** upon dilution for both strong and weak electrolytes (due to increased volume containing 1 mole of electrolyte).
- **Debye-Hückel-Onsager Equation (Strong Electrolytes):** $\Lambda_m = \Lambda^\circ_m - A\sqrt{c}$.

Kohlrausch's Law of Independent Migration of Ions

At infinite dilution ($c \to 0$), the limiting molar conductivity of an electrolyte is the sum of individual contributions of anions and cations:

$$\Lambda^\circ_m = \nu_+\lambda^\circ_+ + \nu_-\lambda^\circ_-$$

Key Applications:

- Limiting molar conductivity of weak electrolyte (e.g., CH_3COOH):

$$\Lambda_m^\circ(CH_3COOH) = \Lambda_m^\circ(CH_3COONa) + \Lambda_m^\circ(HCl) - \Lambda_m^\circ(NaCl)$$

- Degree of Dissociation (α): $\alpha = \frac{\Lambda_m}{\Lambda_m^\circ}$.
- Dissociation Constant (K_a): $K_a = \frac{c\alpha^2}{1-\alpha} = \frac{c\Lambda_m^2}{\Lambda_m^\circ(\Lambda_m^\circ - \Lambda_m)}$.

2.1.3 Faraday's Laws of Electrolysis

- **First Law:** Mass deposited $m \propto Q \implies m = ZQ = ZIt$, where $Z = \frac{E}{96500} = \frac{M}{n \cdot 96500}$.
- **Second Law:** For identical charge passed through series cells: $\frac{m_1}{m_2} = \frac{E_1}{E_2}$.
- 1 Faraday = 96487 $\approx$ 96500 C mol^{-1} = Charge on 1 mole of electrons ($N_A \cdot e$).

2.1.4 Commercial Cells & Batteries

- **Primary Cells:** Reaction cannot be reversed.
  - *Leclanché Dry Cell:* Anode: Zn cylinder; Cathode: Graphite rod surrounded by MnO_2 + C; Electrolyte: $NH_4Cl + ZnCl_2$ paste. Potential $\approx$ 1.5 V.
  - *Mercury Cell:* Anode: Zn–Hg amalgam; Cathode: HgO+C; Electrolyte: KOH+ZnO. Generates constant voltage 1.35 V throughout its life (overall reaction has no ions in solution).
- **Secondary Cells:** Reversible reactions upon recharging.
  - *Lead Storage Battery:* Anode: Spongy Pb; Cathode: PbO_2; Electrolyte: 38% H_2SO_4 (density = 1.30 g/mL).

$$\text{Discharging: } Pb(s) + PbO_2(s) + 2H_2SO_4(aq) \rightarrow 2PbSO_4(s) + 2H_2O(l)$$

    During recharging, the electrical polarity is reversed, regenerating Pb, PbO_2, and H_2SO_4.
- **Fuel Cells (H_2–O_2):** Direct conversion of chemical energy of combustion into electricity. Efficiency $\approx$ 70%, eco-friendly with pure water as byproduct.
- **Corrosion:** Electrochemical mechanism. Anode: $Fe(s) \rightarrow Fe^{2+} + 2e^-$; Cathode: $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$; Rust: $Fe_2O_3 \cdot xH_2O$.

2.2 5. Chemical Kinetics

2.2.1 Rate of Reaction & Rate Law

For reaction $aA + bB \rightarrow cC + dD$:

$$\text{Rate} = -\frac{1}{a}\frac{d[A]}{dt} = -\frac{1}{b}\frac{d[B]}{dt} = +\frac{1}{c}\frac{d[C]}{dt} = +\frac{1}{d}\frac{d[D]}{dt}$$

- **Differential Rate Law:** Rate $= k[A]^x[B]^y$ (Overall order $n = x + y$).
- **Units of Rate Constant k:** $mol^{1-n}\ L^{n-1}\ s^{-1}$ (n = overall reaction order).
- **Order vs Molecularity:**

- Order is experimental, can be zero, fractional, or negative; applies to both elementary and complex reactions.
- Molecularity is theoretical (number of colliding species in an elementary step), strictly an integer $(1, 2, 3)$, never zero or fractional, meaningless for complex overall reactions.

Parameter	**Zero-Order Reaction** ($n = 0$)	**First-Order Reaction** ($n = 1$)	**Second-Order Reaction** ($n = 2$)
Differential Rate	$-\frac{d[A]}{dt} = k$	$-\frac{d[A]}{dt} = k[A]$	$-\frac{d[A]}{dt} = k[A]^2$
Integrated Equation	$[A]_t = [A]_0 - kt$	$[A]_t = [A]_0 e^{-kt} \iff k = \frac{2.303}{t} \log \frac{[A]_0}{[A]_t}$	$\frac{1}{[A]_t} = \frac{1}{[A]_0} + kt$
Linear Plot	$[A]_t$ vs t (Slope: $-k$)	$\log[A]_t$ vs t (Slope: $-\frac{k}{2.303}$)	$\frac{1}{[A]_t}$ vs t (Slope: $+k$)
Half-Life ($t_{1/2}$)	$t_{1/2} = \frac{[A]_0}{2k} \propto [A]_0$	$t_{1/2} = \frac{0.693}{k}$ (**Independent** of $[A]_0$)	$t_{1/2} = \frac{1}{k[A]_0} \propto \frac{1}{[A]_0}$
Completion Time	$t_{100\%} = \frac{[A]_0}{k} = 2t_{1/2}$	$t_{100\%} \to \infty$ (Never 100% complete)	$t_{100\%} \to \infty$
Units of k	mol L^{-1} s^{-1}	s^{-1}	L mol^{-1} s^{-1}

Table 2: Comparison of Kinetic Parameters across Reaction Orders

High-Yield First-Order Proportionality Shortcuts

- $t_{75\%} = 2 \times t_{1/2}$
- $t_{87.5\%} = 3 \times t_{1/2}$
- $t_{93.75\%} = 4 \times t_{1/2}$
- $t_{99\%} \approx \frac{2}{0.301} t_{1/2} \approx 6.64 \times t_{1/2}$
- $t_{99.9\%} \approx 10 \times t_{1/2}$
- Radioactive decay always follows **first-order kinetics**: $N_t = N_0 \left(\frac{1}{2}\right)^n$, where $n = \frac{t}{t_{1/2}}$.

2.2.2 Temperature Dependence & Arrhenius Equation

- **Arrhenius Equation:** $k = Ae^{-E_a/RT}$, where A is the pre-exponential / frequency factor, and E_a is the activation energy.
- **Logarithmic Formulation:**

$$\ln k = \ln A - \frac{E_a}{RT} \iff \log_{10} k = \log_{10} A - \frac{E_a}{2.303RT}$$

  A plot of $\log_{10} k$ versus $\frac{1}{T}$ yields a straight line with Slope $= -\frac{E_a}{2.303R}$ and y-intercept $= \log_{10} A$.
- **Two Temperatures Equation:**

$$\log_{10}\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303R}\left(\frac{T_2 - T_1}{T_1 T_2}\right)$$

- **Temperature Coefficient:** $\frac{k_{T+10}}{k_T} \approx 2$ to 3 for most chemical reactions.
- **Collision Theory:** Rate $= Z_{AB} \cdot P \cdot e^{-E_a/RT}$, where Z_{AB} is collision frequency, and P is the steric / orientation factor.
- **Role of Catalyst:** A catalyst accelerates a reaction by providing an alternative pathway with a **lower activation energy** (E_a). It does **not** alter ΔH, ΔG, or the equilibrium constant (K_{eq}).

2.3 6. Structure of Atom

2.3.1 Bohr's Atomic Model (Single-Electron Species: $\mathbf{H}, \mathbf{He}^+, \mathbf{Li}^{2+}$)

- **Bohr's Quantization Condition:** Angular momentum is quantized in integer multiples of $\hbar$:
$$mvr = \frac{nh}{2\pi} = n\hbar \quad (n = 1, 2, 3, \dots)$$
- **Radius of nth Orbit:**
$$r_n = \frac{n^2 h^2 \varepsilon_0}{\pi m Z e^2} = 0.529 \frac{n^2}{Z}\ \text{Å} = 52.9 \frac{n^2}{Z}\ \text{pm}$$
- **Velocity of Electron in n^{th} Orbit:**
$$v_n = \frac{Ze^2}{2\varepsilon_0 nh} = 2.18 \times 10^6 \frac{Z}{n}\ \text{m/s}$$
- **Total Energy of Electron:**
$$E_n = -13.6 \frac{Z^2}{n^2}\ \text{eV/atom} = -2.18 \times 10^{-18} \frac{Z^2}{n^2}\ \text{J/atom}$$
- **Energy Relations:**
$$K = -E_n = +13.6 \frac{Z^2}{n^2}\ \text{eV}, \qquad U = 2E_n = -27.2 \frac{Z^2}{n^2}\ \text{eV}$$

2.3.2 Rydberg Formula & Hydrogen Emission Spectrum

When an electron de-excites from higher level n_2 to lower level n_1, a photon of wavelength λ is emitted:

$$\bar{\nu} = \frac{1}{\lambda} = R_H Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad (R_H = 109,677\,\text{cm}^{-1} \approx 1.097 \times 10^7\,\text{m}^{-1})$$

- **Lyman Series:** $n_1 = 1, n_2 = 2, 3, 4, \dots \rightarrow$ **Ultraviolet (UV)** region.
- **Balmer Series:** $n_1 = 2, n_2 = 3, 4, 5, \dots \rightarrow$ **Visible** region ($H_\alpha = 656$ nm, red).
- **Paschen Series:** $n_1 = 3, n_2 = 4, 5, \dots \rightarrow$ **Near Infrared (IR)** region.
- **Brackett Series:** $n_1 = 4, n_2 = 5, 6, \dots \rightarrow$ **Infrared (IR)** region.
- **Pfund Series:** $n_1 = 5, n_2 = 6, 7, \dots \rightarrow$ **Far Infrared** region.
- **Humphrey Series:** $n_1 = 6, n_2 = 7, 8, \dots \rightarrow$ **Far Infrared** region.
- **Total Possible Spectral Lines from n^{th} Excited State to Ground State:**
$$N = \frac{n(n-1)}{2}; \qquad \text{Between } n_2 \text{ and } n_1: \quad N = \frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$$

2.3.3 Dual Nature of Matter & Quantum Mechanics

- **de Broglie Wavelength:**
$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mK}} = \frac{h}{\sqrt{2mqV}}$$
For an electron accelerated through potential difference V volts:
$$\lambda_e = \frac{12.27}{\sqrt{V}}\ \text{Å} = \sqrt{\frac{150}{V}}\ \text{Å}$$
- **Heisenberg's Uncertainty Principle:**
$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi} \implies \Delta x \cdot (m\Delta v) \geq \frac{h}{4\pi} \implies \Delta E \cdot \Delta t \geq \frac{h}{4\pi}$$

2.3.4 Quantum Numbers & Electronic Configuration Rules

- **Principal Quantum Number ($n = 1, 2, 3, \ldots$):** Determines main shell size, energy, and max electrons ($2n^2$).
- **Azimuthal Quantum Number ($l = 0$ to $n - 1$):** Subshell shape ($s : 0, p : 1, d : 2, f : 3$). Orbital angular momentum:

$$L = \sqrt{l(l+1)}\frac{h}{2\pi} = \sqrt{l(l+1)}\hbar \quad (\text{For } s\text{-orbital, } l = 0 \implies L = 0)$$

- **Magnetic Quantum Number ($m_l = -l$ to $+l$):** Spatial orientation; total orbitals in subshell $= 2l + 1$.
- **Spin Quantum Number ($m_s = +\frac{1}{2}, -\frac{1}{2}$):** Spin direction. Spin magnetic moment:

$$\mu_s = \sqrt{n(n+2)}\,\text{BM} \quad (\text{Bohr Magneton}, n = \text{number of unpaired electrons})$$

- **Nodes in Orbitals:** Radial nodes $= n - l - 1$; Angular nodes $= l$; Total nodes $= n - 1$.
- **Aufbau Principle:** Orbitals filled in increasing order of $(n + l)$ energy. If $(n + l)$ identical, lower n filled first.
- **Pauli's Exclusion Principle:** No two electrons in an atom can have the exact same set of all four quantum numbers.
- **Hund's Rule of Maximum Multiplicity:** Pairing in degenerate orbitals occurs only after each orbital is singly occupied with parallel spin.

3. Biology — Botany: Plant Diversity & Floral Architecture

3.1 7. Plant Kingdom

3.1.1 Classification Systems

- **Artificial Systems:** Based only on gross superficial morphological characters (habit, colour, number, shape of leaves, androecium structure; Linnaeus).
- **Natural Systems:** Based on natural affinities among organisms, considering internal features like ultrastructure, anatomy, embryology, and phytochemistry (George Bentham & Joseph Dalton Hooker).
- **Phylogenetic Systems:** Based on evolutionary relationships; organisms in the same taxa share a common ancestor (Engler & Prantl, Hutchinson).
- **Modern Taxonomical Aids:** Numerical taxonomy (computers, equal weightage to all characters), Cytotaxonomy (chromosome number, structure, behavior), Chemotaxonomy (chemical constituents).

3.1.2 Algae Division Breakdown

3.1.3 Bryophytes (Amphibians of the Plant Kingdom)

- Live in soil but **depend on water for sexual reproduction** (ciliated male antherozoids swim to archegonium).
- **Plant Body:** Thalloid, attached by unicellular or multicellular rhizoids. Lacks true roots, stems, and leaves.

Feature	Chlorophyceae (Green)	Phaeophyceae (Brown)	Rhodophyceae (Red)
Major Pigments	Chlorophyll *a*, *b*	Chlorophyll *a*, *c*, Fucoxanthin	Chlorophyll *a*, *d*, *r*-Phycoerythrin
Stored Food	Starch (in pyrenoids)	Laminarin and Mannitol	Floridean starch (like amylopectin)
Cell Wall	Cellulose (inner) + Pectose (outer)	Cellulose + gelatinous Algin	Cellulose, Pectin, Polysulphate esters
Flagella	2–8, equal, apical	2, unequal, lateral	**Absent** in all stages (non-motile)
Habitat	Fresh water, brackish, salt water	Mostly marine (rare fresh water)	Mostly marine, warm coastal waters
Examples	*Chlamydomonas*, *Volvox*, *Ulothrix*, *Spirogyra*, *Chara*	*Ectocarpus*, *Dictyota*, *Laminaria*, *Sargassum*, *Fucus*	*Polysiphonia*, *Porphyra*, *Gracilaria*, *Gelidium* (Agar)

Table 3: Comparative Overview of Algal Classes (NCERT Canonical)

- **Dominant Phase:** Free-living, photosynthetic, haploid **Gametophyte** (n).
- **Sporophyte ($2n$):** Differentiated into foot, seta, and capsule. Not free-living; permanently attached to and nutritionally dependent on the gametophyte.
- **Liverworts:** E.g., *Marchantia*. Dorsiventral thallus; asexual reproduction via **gemma cups** containing multicellular green buds called **gemmae**. Dioecious (male antheridiophore, female archegoniophore).
- **Mosses:** Two distinct developmental stages:
  - *Protonema Stage:* Creeping, green, branched, filamentous directly from spore.
  - *Leafy Stage:* Arises as lateral bud from secondary protonema; has spirally arranged leaves.
  - Capsule possesses **peristome teeth** mechanism for regulated spore dispersal.
  - Examples: *Funaria*, *Polytrichum*, *Sphagnum* (Peat moss: used as packing material for transshipment of living organisms due to high water-holding capacity; fuel source).

3.1.4 Pteridophytes (First Vascular Land Plants)

- Possess xylem (tracheids, lack vessels) and phloem (sieve cells, lack companion cells).
- **Dominant Phase:** Independent diploid **Sporophyte** ($2n$) with true roots, stem, and leaves.
- Leaves may be small microphylls (*Selaginella*) or large macrophylls (Ferns).
- Sporangia subtended by leaf-like appendages called **sporophylls**, forming cones/strobili (*Selaginella*, *Equisetum*).
- Spores germinate into **Prothallus** (inconspicuous, small, multicellular, green, free-living haploid gametophyte requiring cool, damp conditions).
- **Heterospory & Seed Habit Precursor:** Most pteridophytes are homosporous. ***Selaginella*** and ***Salvinia*** are heterosporous (produce microspores and megaspores). Megaspore retention within parent sporophyte and development of female gametophyte on sporophyte represents a pivotal evolutionary step toward the **seed habit**.
- Classes: Psilopsida (*Psilotum*), Lycopsida (*Selaginella*, *Lycopodium*), Sphenopsida (*Equisetum*), Pteropsida (*Dryopteris*, *Pteris*, *Adiantum*).

3.1.5 Gymnosperms & Angiosperms

- **Gymnosperms (Naked Seeds):** Ovules are not enclosed by any ovary wall and remain exposed both before and after fertilization. Seeds lack fruit covering.

- *Pinus:* Monoecious; branched stem; mycorrhizal symbiotic roots; winged pollen; needle-like leaves with sunken stomata and thick cuticle.
- *Cycas:* Dioecious; unbranched stem; coralloid roots with nitrogen-fixing cyanobacteria (*Anabaena*, *Nostoc*); motile ciliated male gametes.
- Endosperm is **haploid (n)** and formed **before fertilization**.

- **Angiosperms:** Ovules enclosed within ovary; seeds enclosed inside fruit.
- **Double Fertilization (Unique to Angiosperms):**

$$\text{Syngamy: Egg } (n) + \text{Male Gamete } (n) \rightarrow \text{Zygote } (2n)$$

$$\text{Triple Fusion: Polar Nuclei } (2n) + \text{Male Gamete } (n) \rightarrow \text{Primary Endosperm Nucleus (PEN) } (3n)$$

Plant Life Cycle Patterns

- **Haplontic:** Gametophyte (n) dominant; sporophyte represented only by single-celled zygote ($2n$) which undergoes meiosis. Examples: *Volvox*, *Spirogyra*, *Chlamydomonas*.
- **Diplontic:** Sporophyte ($2n$) dominant, photosynthetic, independent; gametophyte reduced to few cells. Examples: All Gymnosperms, Angiosperms, and the brown alga *Fucus*.
- **Haplodiplontic:** Both phases multicellular and intermediate. Examples: Bryophytes, Pteridophytes, and algae *Ectocarpus*, *Polysiphonia*, Kelps.

3.2 8. Morphology of Flowering Plants

3.2.1 Root Modifications

- **Regions of Root Tip:** Root cap → Meristematic zone (rapid mitosis) → Elongation zone (responsible for root growth in length) → Maturation zone (bears root hairs).
- **Storage Roots:** Tap roots in Carrot, Turnip, Beetroot; Adventitious roots in Sweet potato (*Ipomoea batatas*).
- **Prop Roots (Pillar-like support):** Arise from horizontal branches of Banyan (*Ficus benghalensis*).
- **Stilt Roots:** Arise from lower nodes of stem for support in Maize and Sugarcane.
- **Pneumatophores (Respiratory roots):** Grow vertically upward from swampy soil in halophytes like *Rhizophora* to obtain oxygen.

3.2.2 Stem Modifications

- **Underground Food Storage & Perennation:** Rhizome (Ginger, Turmeric); Tuber (Potato); Corm (*Colocasia*, *Amorphophallus* / Zaminkand); Bulb (Onion, Garlic).
- **Stem Tendrils (Axillary buds):** Slender, spirally coiled for climbing in Gourds (Cucumber, Pumpkin, Watermelon) and Grapevines.
- **Thorns (Axillary buds):** Woody, straight, pointed for defense against browsers in *Citrus* and *Bougainvillea*.
- **Phylloclades (Photosynthetic stems):** Flattened green stem in *Opuntia*; fleshy cylindrical in *Euphorbia*. Cladodes: one internode long in *Asparagus*.
- **Vegetative Propagation:** Runners (Grasses, Strawberry); Stolons (Mint, Jasmine); Offsets (one-internode lateral branch with rosette of leaves: *Pistia*, *Eichhornia*); Suckers (Banana, Pineapple, *Chrysanthemum*).

3.2.3 Leaf Morphology & Inflorescence

- **Venation:** Reticulate (network, Dicots) vs Parallel (parallel veins, Monocots).
- **Phyllotaxy (Arrangement of leaves on stem):**
  - Alternate: Single leaf at each node (Mustard, China rose, Sunflower).
  - Opposite: Pair of leaves at each node (*Calotropis*, Guava).
  - Whorled: More than two leaves at a node (*Alstonia*).
- **Leaf Modifications:** Tendrils (Pea); Spines (Cactus); Phyllode (expanded green petiole: Australian *Acacia*); Pitcher (*Nepenthes*).
- **Inflorescence:**
  - Racemose: Main axis continues to grow indefinitely; flowers borne laterally in **acropetal succession** (oldest at base).
  - Cymose: Main axis terminates in a flower (limited growth); flowers borne in **basipetal succession** (oldest at apex).

3.2.4 Flower Anatomy: Symmetry, Ovary Position & Aestivation

- **Symmetry:** Actinomorphic (radial: Mustard, *Datura*, Chilli); Zygomorphic (bilateral: Pea, Gulmohar, Bean, *Cassia*); Asymmetric (irregular: *Canna*).
- **Ovary Insertion:**
  - Hypogynous (Superior Ovary): Mustard, China rose, Brinjal.
  - Perigynous (Half-Inferior Ovary): Thalamus cup-shaped; Plum, Rose, Peach.
  - Epigynous (Inferior Ovary): Thalamus completely encloses ovary; Guava, Cucumber, ray florets of Sunflower.
- **Aestivation (Petal/Sepal arrangement in bud):**
  - Valvate: Margins touch without overlapping (*Calotropis*).
  - Twisted: One margin regularly overlaps the next (China rose, Lady's finger, Cotton).
  - Imbricate: Margins overlap irregularly with no direction (*Cassia*, Gulmohar).
  - Vexillary (Papilionaceous): 5 petals: 1 large posterior Standard (Vexillum), 2 lateral Wings (Alae), 2 anterior fused Keel (Carina). Examples: Pea, Bean.
- **Placentation:**
  - Marginal: Ovules along ventral suture of unilocular ovary (Pea).
  - Axile: Multilocular ovary, placentae on central axis (China rose, Tomato, Lemon).
  - Parietal: Ovules on inner wall; 1-chambered becoming 2-chambered via false septum (replum) (Mustard, *Argemone*).
  - Free Central: Ovules on central axis, septa absent (Dianthus, Primrose).
  - Basal: Single ovule at base of unilocular ovary (Sunflower, Marigold).

Feature	Fabaceae (Legume / Pea)	Solanaceae (Potato Family)	Liliaceae (Lily / Monocot)
Habit	Herbs, shrubs, trees; root nodules	Herbs, shrubs, rarely small trees	Perennial herbs with bulb/corm
Leaves	Pinnately compound, pulvinate base	Alternate, simple, exstipulate	Basal, alternate, parallel venation
Symmetry	Zygomorphic (%)	Actinomorphic ($\oplus$)	Actinomorphic ($\oplus$)
Calyx / Corolla	$K_{(5)}$; $C_{1+2+(2)}$ (Vexillary)	$K_{(5)}$ (persistent); $C_{(5)}$ (valvate)	Perianth: $P_{(3+3)}$ (tepals united)
Androecium	$A_{(9)+1}$ (Diadelphous, dithecous)	A_5 (Epipetalous, alternipetalous)	A_{3+3} (Epitepalous)
Gynoecium	$\underline{G}_1$ (Monocarpellary, superior)	$\underline{G}_{(2)}$ (Bicarpellary, oblique, swollen)	$\underline{G}_{(3)}$ (Tricarpellary, trilocular)
Placentation	Marginal placentation	Axile placentation (swollen placenta)	Axile placentation
Fruit & Seed	Legume; Non-endospermic	Berry or capsule; Endospermic	Capsule; Endospermic
Floral Formula	% $K_{(5)}$ $C_{1+2+(2)}$ $A_{(9)+1}$ $\underline{G}_1$	$\oplus$ $K_{(5)}$ $\widehat{C}_{(5)}$ A_5 $\underline{G}_{(2)}$	$\oplus$ $\widehat{P}_{(3+3)}$ A_{3+3} $\underline{G}_{(3)}$
Examples	Gram, Arhar, Moong, Soya, Groundnut, Mulethi, Indigofera	Tomato, Brinjal, Potato, Chilli, Belladonna, Ashwagandha, Petunia	Tulip, *Gloriosa*, *Aloe*, *Asparagus*, *Colchicum autumnale*

Table 4: Comprehensive Comparison of NEET Syllabus Angiosperm Families

3.2.5 Key Angiosperm Families

4. Biology — Zoology: Excretory Systems & Musculoskeletal Mechanics

4.1 9. Excretory Products and Their Elimination

4.1.1 Nitrogenous Waste Excretion Modes

- **Ammonotelism:** Excretion of highly toxic ammonia (NH_3). Highly soluble, requires massive water volume (300–500 mL/g). Diffuses readily across body/gill surfaces as ammonium ions (NH_4^+). Kidneys play **no significant role** in ammonia excretion. Examples: Bony fishes, aquatic amphibians (tadpole), aquatic insects.
- **Ureotelism:** Excretion of urea. Ammonia converted to urea in the **liver** via the **Ornithine (Urea) Cycle**. Moderately toxic, requires less water (50 mL/g). Some urea retained in medullary matrix to maintain interstitial hyperosmolarity. Examples: Mammals, terrestrial amphibians (adult frog), cartilaginous marine fishes (sharks).
- **Uricotelism:** Excretion of uric acid as a white paste or insoluble pellet with minimal water loss (10 mL/g). Least toxic mode, vital terrestrial water conservation adaptation. Examples: Reptiles, birds, land snails, insects.

4.1.2 Human Excretory System & Nephron Architecture

- **Kidneys:** Retroperitoneal, bean-shaped, situated between T_{12} and L_3 vertebrae. Longitudinal section shows outer cortex and inner medulla organized into renal medullary pyramids separated by **Columns of Bertini**.
- **Nephron:** Structural and functional unit (≈ 1 million per kidney).

- *Renal Corpuscle (Malpighian Body):* Glomerulus (capillary tuft from afferent arteriole, draining into narrower efferent arteriole) + Bowman's Capsule. The inner visceral layer of Bowman's capsule is lined by specialized **podocytes** with intricate pedicels leaving filtration slits (slit pores).
- *Renal Tubule:* Proximal Convoluted Tubule (PCT) → Henle's Loop (descending limb + ascending limb) → Distal Convoluted Tubule (DCT) → Collecting Duct.
- *Cortical Nephrons (85%):* Short loop of Henle extending barely into medulla; vasa recta highly reduced or absent.
- *Juxtamedullary Nephrons (15%):* Deep loop of Henle descending into inner medulla; well-developed vasa recta; indispensable for **concentrating urine**.

4.1.3 Urine Formation Mechanisms

- **1. Glomerular Ultrafiltration:** Filtration across three layers: glomerular capillary endothelium, basement membrane, and podocyte slit pores.

$$\text{Net Filtration Pressure (NFP)} = \text{GHP} - (\text{BCOP} + \text{CHP}) = 60 - (32 + 18) = 10\,\text{mmHg}$$

  Glomerular Filtration Rate (GFR): 125 mL/min = 180 L/day.

- **2. Tubular Reabsorption:** Over 99% of filtrate is reabsorbed (180 L filtrate → 1.5 L urine daily).
  - *PCT:* Reabsorbs 70–80% of electrolytes and water. Reabsorbs 100% of glucose and amino acids by secondary active transport. Lined by simple cuboidal epithelium with **brush border microvilli**.
  - *Henle's Loop:*
    - *Descending Limb:* Permeable to water, virtually impermeable to electrolytes $\implies$ filtrate becomes concentrated (**hypertonic**).
    - *Ascending Limb:* Impermeable to water, actively and passively transports electrolytes (Na^+, Cl^-) into interstitium $\implies$ filtrate becomes dilute (**hypotonic**).
  - *DCT:* Conditional reabsorption of Na^+ and water under Aldosterone and ADH. Reabsorbs HCO_3^-.
  - *Collecting Duct:* Reabsorbs substantial water to yield concentrated urine; diffuses small quantities of urea into medullary interstitium.

- **3. Tubular Secretion:** Active secretion of H^+, K^+, and NH_3 into the filtrate in PCT and DCT to maintain systemic ionic balance and physiological pH (7.4).

Counter-Current Multiplier System

Components: Loop of Henle and Vasa Recta running in antiparallel counter-current flows.

- Establishes a steep medullary osmotic gradient: 300 mOsmol/L in renal cortex up to 1200 mOsmol/L in deep inner medulla.
- Mediated by active transport of $NaCl$ from ascending Henle's limb into descending vasa recta, and diffusion of urea from collecting duct.
- Permits human kidneys to produce urine **4 times more concentrated** ($\approx$ 1200 mOsmol/L) than initial blood filtrate (300 mOsmol/L).

4.1.4 Hormonal Regulation of Kidney Function

- **Hypothalamic Osmoreceptor–ADH Axis:** Dehydration or high osmolarity stimulates hypothalamic osmoreceptors → Neurohypophysis releases **Antidiuretic Hormone (ADH / Vasopressin)** → Inserts aquaporin channels into DCT and Collecting Duct → Massive water reabsorption (prevents diuresis) + systemic vasoconstriction elevating BP and GFR.
- **Renin–Angiotensin–Aldosterone System (RAAS):** Fall in glomerular blood flow/pressure stimulates **Juxtaglomerular (JG) cells** to secrete **Renin**:

$$\text{Angiotensinogen (Liver)} \xrightarrow{\text{Renin}} \text{Angiotensin I} \xrightarrow{\text{ACE (Lungs)}} \text{Angiotensin II}$$

  Angiotensin II is a potent vasoconstrictor and stimulates adrenal cortex to release **Aldosterone**, promoting active Na^+ and water reabsorption in DCT.
- **Atrial Natriuretic Factor (ANF):** High venous return/BP stretches heart atria → releases ANF → Vasodilation + promotes Na^+ excretion (natriuresis) → Lowers BP, directly **antagonizing and checking the RAAS axis**.

4.1.5 Excretory Disorders

- **Uremia:** Accumulation of urea and nitrogenous toxins in blood due to kidney failure. Managed by **hemodialysis**: blood drawn from radial artery, mixed with anticoagulant (heparin), passed through dialyzing unit with cellophane tubes immersed in dialyzing fluid (identical to plasma except lacking nitrogenous wastes), then cleared blood mixed with anti-heparin and returned through a vein.
- **Renal Calculi (Kidney Stones):** Insoluble crystallized salts, primarily **calcium oxalate**, precipitating in renal pelvis.
- **Glomerulonephritis:** Inflammation of glomeruli caused by immune complex deposition.

4.2 10. Locomotion and Movement

4.2.1 Types of Movement & Muscle Tissue Classification

- **Types of Animal Movement:** Amoeboid (pseudopodia with actin microfilaments: Macrophages and Neutrophils); Ciliary (coordinated cilia: respiratory trachea, fallopian tubes/oviducts); Flagellar (spermatozoa); Muscular (limbs, jaw, tongue).
- **Skeletal Muscle:** Striated, voluntary, multinucleated (**syncytium**), peripherally positioned nuclei, attached to bones by tendons.
- **Visceral / Smooth Muscle:** Non-striated, involuntary, uninucleated, spindle-shaped (fusiform), located in walls of hollow internal visceral organs (gut, blood vessels, bladder).
- **Cardiac Muscle:** Striated, involuntary, uninucleated, branched, with **intercalated discs** containing gap junctions allowing synchronized electrical contraction.

4.2.2 Skeletal Muscle Ultrastructure

- **Organizational Hierarchy:** Skeletal Muscle → Muscle Fascicles (enclosed in Perimysium) → Muscle Fibres / Cells (enclosed in Sarcolemma) → Myofibrils → Myofilaments.
- **Sarcoplasmic Reticulum (SR):** Specialized endoplasmic reticulum acting as intracellular storehouse for calcium ions (Ca^{2+}).
- **Sarcomere (Functional Contractile Unit):** Segment between two adjacent **Z-lines**.

- **A-Band (Anisotropic):** Dark band; contains thick myosin filaments overlapping with thin actin filaments; length remains **constant** during contraction.
- **I-Band (Isotropic):** Light band; contains thin actin filaments only; bisected by the elastic Z-line.
- **H-Zone:** Central light zone of A-band where thick filaments are **not** overlapped by thin filaments.
- **M-Line:** Thin fibrous membrane running through the middle of the H-zone holding thick filaments.

Molecular Architecture of Contractile Proteins

1. Thin Filament (Actin):

- Two helically wound strands of filamentous **F-actin**, each a polymer of monomeric globular **G-actin**.
- Two strands of **Tropomyosin** run along the grooves of F-actin throughout its length.
- **Troponin Complex:** Globular protein distributed at regular intervals on tropomyosin:
  - *Troponin T:* Binds to tropomyosin.
  - *Troponin I:* Inhibits actin–myosin interaction by masking active sites on actin.
  - *Troponin C:* Binds to Ca^{2+} ions during excitation.

2. Thick Filament (Myosin):

- Polymer of monomeric **Meromyosins**.
- **Heavy Meromyosin (HMM):** Globular head + short arm; projects outward as cross-arm; contains active **ATPase enzyme** and specific **Actin-binding sites**.
- **Light Meromyosin (LMM):** Elongated tail segment.

4.2.3 Sliding Filament Theory of Muscle Contraction

- **Step 1: Neuromuscular Transmission:** Motor neuron action potential reaches motor end plate → Acetylcholine (ACh) released into synaptic cleft → binds sarcolemma receptors → Action potential propagates along sarcolemma into transverse T-tubules.
- **Step 2: Ca^{2+} Release & Unmasking:** Action potential triggers sarcoplasmic reticulum to release Ca^{2+} into sarcoplasm. Ca^{2+} binds **Troponin C** → conformational shift in tropomyosin → unmasks active myosin-binding sites on actin.
- **Step 3: Cross-Bridge Formation:** Myosin head hydrolyzes ATP ($ATP \rightarrow ADP + P_i$) via ATPase; energized head binds unmasked actin active site forming a **cross-bridge**.
- **Step 4: Power Stroke:** Myosin head tilts and pulls actin filaments toward the centre of the sarcomere (M-line), releasing ADP and P_i.
- **Structural Changes During Contraction:**
  - Sarcomere shortens (Z-lines pulled closer).
  - **I-band reduces** in length.
  - **H-zone narrows and disappears** upon full contraction.
  - **A-band remains strictly constant in length** (Crucial NEET fact!).

- **Step 5: Cross-Bridge Detachment:** A new ATP molecule binds to the myosin head, breaking the cross-bridge. ATP hydrolysis recocks the head for the next cycle.
- *Rigor Mortis:* Post-mortem ATP depletion prevents detachment of cross-bridges, causing sustained muscular rigidity.

4.2.4 Red vs White Muscle Fibres

Characteristic	Red Muscle Fibres (Slow-Twitch)	White Muscle Fibres (Fast-Twitch)
Myoglobin Content	High (deep red color)	Low / trace (pale or whitish color)
Mitochondria	Abundant (rich aerobic machinery)	Few
Sarcoplasmic Reticulum	Moderate / less extensive	Highly extensive (rapid Ca^{2+} surge)
Contraction Velocity	Slower, sustained contraction	Fast, explosive contraction
Fatigue Resistance	High resistance to fatigue; aerobic	Rapid fatigue due to lactic acid accumulation
Examples	Extensor muscles of human back, flight muscles of migratory birds	Eyeball muscles, flight muscles of sparrows

Table 5: Comparison of Skeletal Muscle Fibre Types

4.2.5 Human Skeletal System (Total 206 Bones)

- **Axial Skeleton (80 Bones):**
  - **Skull (22 bones):**
    - *Cranial Bones (8):* Frontal (1), Parietal (2), Temporal (2), Occipital (1 with 2 occipital condyles → **dicondylic skull**), Sphenoid (1), Ethmoid (1).
    - *Facial Bones (14):* Nasal (2), Maxillae (2), Zygomatic (2), Lacrimal (2), Palatine (2), Inferior nasal conchae (2), Vomer (1), Mandible (1, only movable skull bone).
  - **Associated Skull Bones (7):** Hyoid bone (1, U-shaped at buccal cavity base); Ear ossicles (6: Malleus 2, Incus 2, Stapes 2 — smallest bone in human body).
  - **Vertebral Column (26 bones):** Cervical (C_7, C_1 = Atlas, C_2 = Axis), Thoracic (T_{12}), Lumbar (L_5), Sacral ($S_{(1)}$ fused from 5), Coccygeal ($Co_{(1)}$ fused from 4).
  - **Sternum (1):** Flat breastbone on ventral midline of thorax.
  - **Ribs (12 pairs = 24 bones):** Bicephalic dorsal articulation.
    - *True Ribs (Pairs 1–7):* Vertebrosternal; attach dorsally to vertebrae and ventrally to sternum via hyaline costal cartilage.
    - *False Ribs (Pairs 8–10):* Vertebrochondral; attach ventrally to the costal cartilage of the 7th rib instead of directly to the sternum.
    - *Floating Ribs (Pairs 11–12):* Vertebral; not connected ventrally to the sternum at all.
- **Appendicular Skeleton (126 Bones):**
  - **Pectoral Girdle (4 bones):** Scapula (2, flat triangular bone between 2nd and 7th ribs, dorsal spine terminating in acromion process, glenoid cavity articulating with humerus) + Clavicle (2, collar bone, S-shaped with two curvatures).
  - **Upper Limb Bones (60 bones = 30 × 2):** Humerus (1), Radius (1), Ulna (1), Carpals (8 wrist bones), Metacarpals (5 palm bones), Phalanges (14: formula $2, 3, 3, 3, 3$).
  - **Pelvic Girdle (2 bones):** Two coxal (hip) bones. Each formed by fusion of **Ilium, Ischium, and Pubis**. At their junction is the cup-shaped **Acetabulum** articulating with the femoral head. Two pubic bones join ventrally at the **Pubic Symphysis** (fibrocartilage).

- **Lower Limb Bones (60 bones = 30 × 2):** Femur (1, longest and strongest bone), Patella (1, sesamoid kneecap), Tibia (1), Fibula (1), Tarsals (7 ankle bones), Metatarsals (5 sole bones), Phalanges (14: formula $2, 3, 3, 3, 3$).

4.2.6 Joint Classifications & Musculoskeletal Disorders

- **Fibrous Joints (Synarthroses):** Immovable; bones firmly joined by dense fibrous connective tissue. Example: Sutures between skull bones.
- **Cartilaginous Joints (Amphiarthroses):** Slightly movable; bones joined by cartilage pads. Examples: Intervertebral discs between adjacent vertebrae, pubic symphysis.
- **Synovial Joints (Diarthroses):** Freely movable; characterized by a fluid-filled **synovial cavity** lined by synovial membrane between articulating cartilage-covered ends.
  - *Ball and Socket Joint:* Shoulder joint (humerus & glenoid cavity) and Hip joint (femur & acetabulum).
  - *Hinge Joint:* Knee joint, elbow joint, interphalangeal joints.
  - *Pivot Joint:* Between Atlas and Axis (atlanto-axial joint; "no" head movement); proximal radioulnar joint.
  - *Gliding Joint:* Between adjacent carpal bones of wrist; between tarsals of ankle.
  - *Saddle Joint:* Between Carpal and Metacarpal of the thumb (**Classic NEET question!**).
  - *Condyloid / Ellipsoid Joint:* Radiocarpal wrist joint; atlanto-occipital joint ("yes" nodding movement).
- **Disorders:**
  - **Myasthenia Gravis:** Autoimmune disorder affecting neuromuscular junction; antibodies attack acetylcholine receptors; leads to progressive fatigue, weakening, and paralysis of skeletal muscles.
  - **Tetany:** Rapid, violent involuntary muscle spasms caused by hypocalcemia (low Ca^{2+} in body fluids).
  - **Muscular Dystrophy:** Progressive genetic degeneration of skeletal muscle (e.g., Duchenne muscular dystrophy, X-linked recessive defect in dystrophin).
  - **Osteoarthritis:** Age-related degeneration of articular cartilage, causing joint friction, pain, and bone spurs.
  - **Rheumatoid Arthritis:** Autoimmune inflammation of synovial membrane (pannus formation), eroding articular cartilage; causes severe pain, swelling, and joint deformity.
  - **Gout (Gouty Arthritis):** Metabolic disease caused by accumulation and crystallization of **uric acid** in synovial joints, particularly the big toe.
  - **Osteoporosis:** Age-related reduction in bone mineral density and deterioration of bone microarchitecture, increasing fracture risk; commonly triggered by **estrogen deficiency** in postmenopausal women.

End of NEET High-Yield Revision Guide | Prepared by spdedsec